



DPUAxControlLUT

Acceleration controller based on lookup tables.

1. Inputs:

<i>Activate</i>	= 0 : controller is not active, i.e. the inputs GasPedalIn and BrakeIn are passed through to the outputs GasPedalOut and BrakeOut respectively. = 1 : controller is active, i.e. the outputs GasPedalOut and BrakeOut are calculated to reach the desired acceleration of axTarget.
<i>GasPedalIn</i>	acceleration pedal from mockup (driver's input)
<i>BrakeIn</i>	brake pedal from mockup (driver's input)
<i>axTarget</i>	desired acceleration (positive value) or deceleration (negative value) [m/s^2]
<i>v</i>	current velocity of vehicle [m/s].
<i>PitchRoad</i>	pitch of the road [rad]

2. Dynamic parameters:

<i>axTargetMin</i>	minimal acceleration (maximal deceleration) to be reached [m/s^2] default: -1000.0
<i>axTargetMax</i>	maximal acceleration to be reached [m/s^2] default: 1000.0
<i>axTargetDotMin</i>	maximal change rate of deceleration [m/s^3] default: -1000.0
<i>axTargetDotMax</i>	maximal change rate of acceleration [m/s^3] default: 1000.0
<i>epsInterpolate</i>	if no suitable value for deceleration can be found in the lookup table with a deviation less than this value, an inexact interpolation is used default: 0.3
<i>axFactor</i>	factor for interpolation for ax default: 0.25
<i>vFactor</i>	factor for interpolation for v default: 1.0
<i>Offset</i>	offset value for interpolation default: 0.15

3. Static parameters:

<i>LUTAccel</i>	Path to the lookup table file with acceleration values (see functionality)
<i>LUTDecel</i>	Path to the lookup table file with deceleration values (see functionality)



4. Outputs:

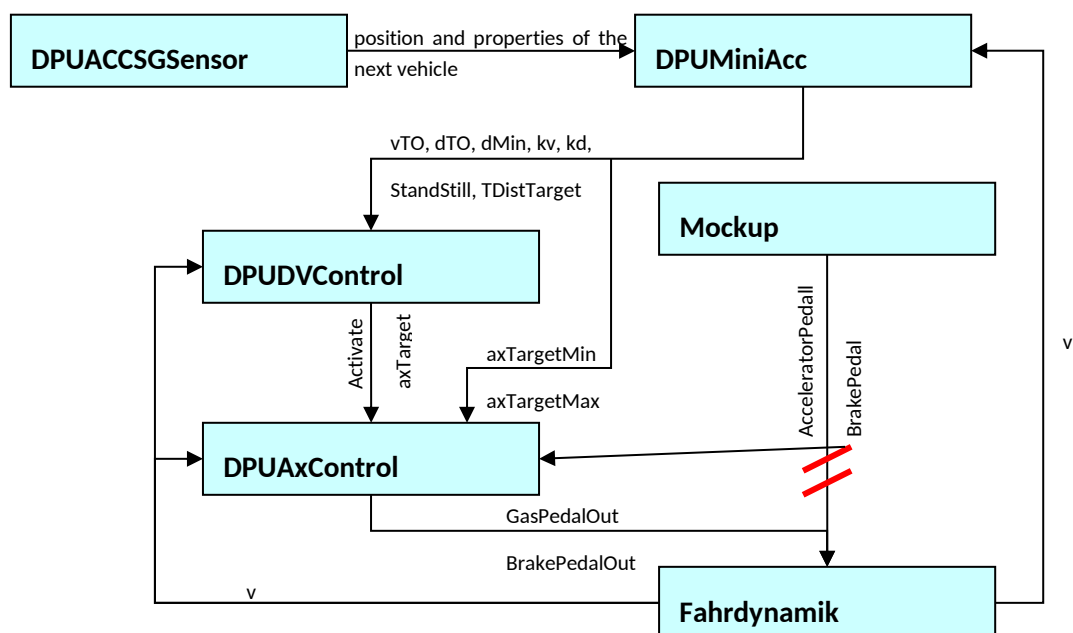
GasPedalOut accelerator pedal

BrakeOut brake pedal

axTargetOut limited acceleration/deceleration because of the dynamic parameters

5. Functional principle:

DPUAxControlLUT is interconnected between mockup and vehicle dynamics of the simulator. The signals of accelerator pedal and brake pedal are connected with the inputs GasPedalIn and BrakePedalIn accordingly. The outputs GasPedalOut and BrakePedalOut are connected with the respective inputs of the vehicle dynamic simulation. When Activate is set to 1, the inputs from the mockup are ignored and the output values are calculated to match the desired acceleration. Otherwise when Activate is set to 0 (i.e. the controller is not active), the signals are just passed through without any alternation.



The controller is based on two lookup table files, one for the acceleration ($axTarget > 0$), the other for deceleration ($axTarget < 0$). For a correct function of the controller these files must be created with the vehicle dynamics and parameters used in the simulation. The SILAB package contains lookup table files for the vehicle dynamic simulation DPUMstVeda and BMW520 configuration.

Every lookup table file is a matrix. The columns correlate to velocity values, the rows contain acceleration or brake pedal values. A positive value equals to a acceleration pedal value while brake pedal values are encoded as a negative value. Every line contains an acceleration value which can be reached with the corresponding pedal value. The matrix is stored in an ASCII file with the following format:



```
<number of columns (int)>
<number of rows (int)>
<minimal pedal value (first line of the matrix)>
<maximal pedal value (last line of the matrix)>
<maximum velocity (last column of the matrix) [km/h]>
<accuracy to search for velocity [km/h]>
<accuracy to search for acceleration [m/s²]>
<line 1, column 1: velocity [km/h]>
<line 1, column 1: pedal value>
<line 1, column 1: acceleration [m/s²]>
<line 1, column 2: velocity [km/h]>
<line 1, column 2: pedal value>
<line 1, column 2: acceleration [m/s²]>
.
.
.
<line 2, column 1: velocity [km/h]>
<line 2, column 1: pedal value>
<line 2, column 1: acceleration [m/s²]>
.
.
.
```

In case the lookup table isn't suitable for high deceleration values the brake pedal position can be calculated by the following formula:

$$axFactor * ax - vFactor / (1 + v^2) + Offset$$

Where ax is the desired deceleration and v the current velocity.

6. Example:

```
DPUAxControlLUT AxControl
{
    Computer = {VDYN};
    Index = 203;
    LUTAccel = "mstVedaAutoPar.BMW/Accel_BMW5.lut";
    LUTDecel = "mstVedaAutoPar.BMW/Decel_BMW5.lut";
};

Connections =
{
    # get activation and target acceleration/deceleration
    DVControl.ActivateAxControl -> AxControl.Activate,
    DVControl.axTarget -> AxControl.axTarget,
    # get current velocity from vehicle dynamics
```



SILAB DRIVING SIMULATION

version Version 7.1 documentation

```
VDyn.v -> AxControl.v,  
# get current pitch of the road from scenery  
SCNX.PitchRoad -> AxControl.PitchRoad,  
# driver's inputs from mockup  
Mockup.AcceleratorPedal -> AxControl.GasPedalIn,  
Mockup.BrakePedal -> AxControl.BrakeIn,  
  
# outputs to vehicle dynamics  
AxControl.GasPedalOut -> VDyn.AcceleratorPedal,  
AxControl.BrakeOut -> VDyn.BrakePedal  
};
```